COURSE: CSA1613
DataWarehousing and Data Mining

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EXPERIMENT 1: ADDITION

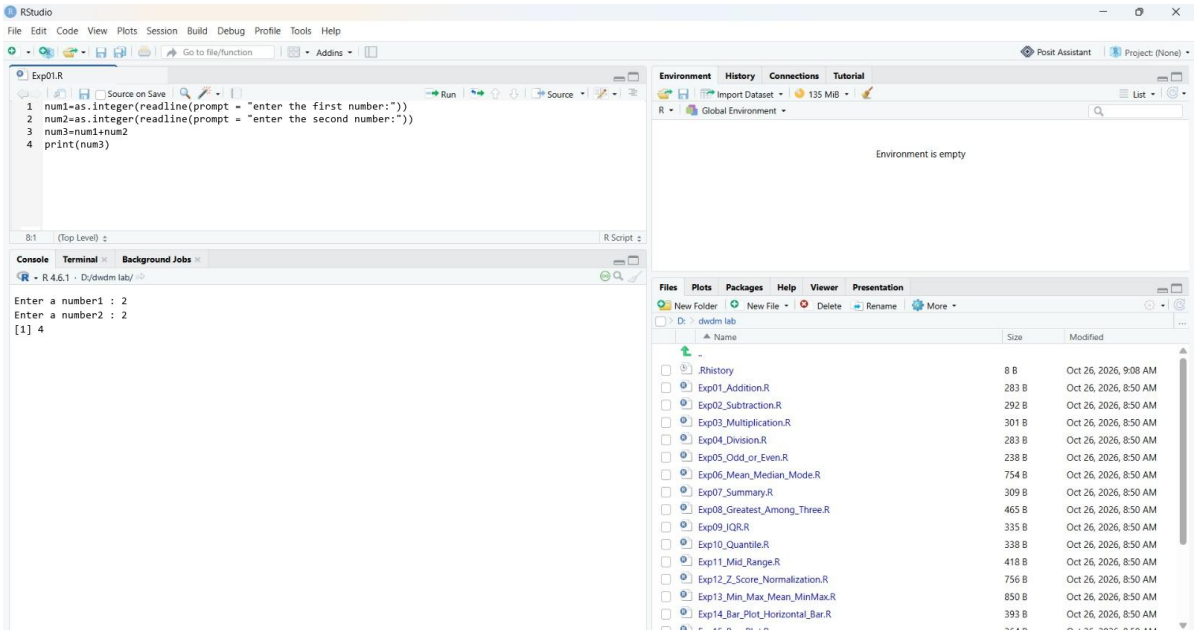
Aim

To prove the program for addition using R-tool.

Program1

```
num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
num3=num1+num2
print(num3)
```

Output



Result

Thus the basic program addition is executed successfully.

EXPERIMENT 2: SUBTRACTION

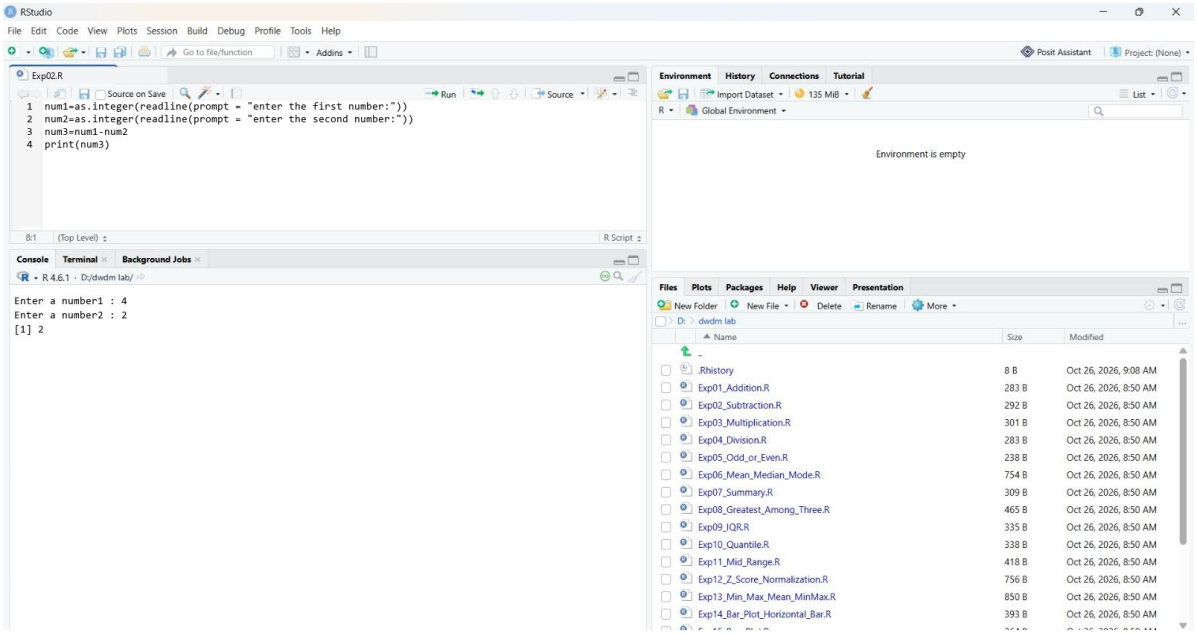
Aim

To prove the program for subtraction using R-tool.

Program

```
num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
num3=num1-num2
print(num3)
```

Output



Result

Thus, the basic program subtraction is executed successfully.

EXPERIMENT 3: MULTIPLICATION

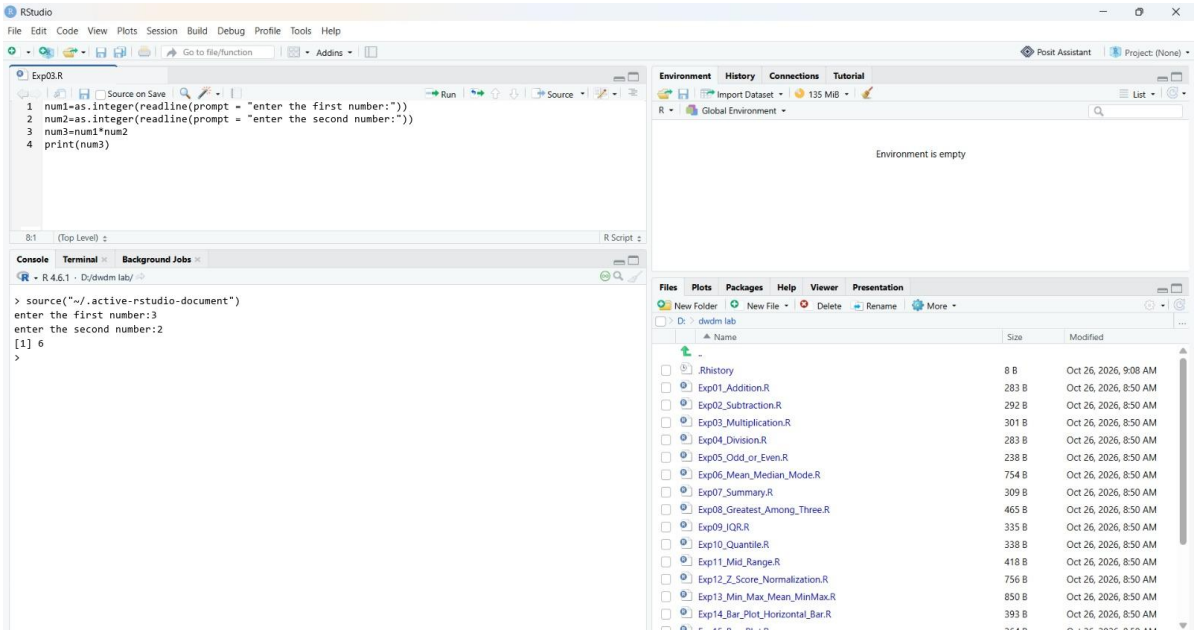
Aim

To prove the program for multiplication using R-tool.

Program

```
num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
num3=num1*num2
print(num3)
```

Output



Result

Thus, the basic program multiplication is executed successfully.

EXPERIMENT 4: DIVISION

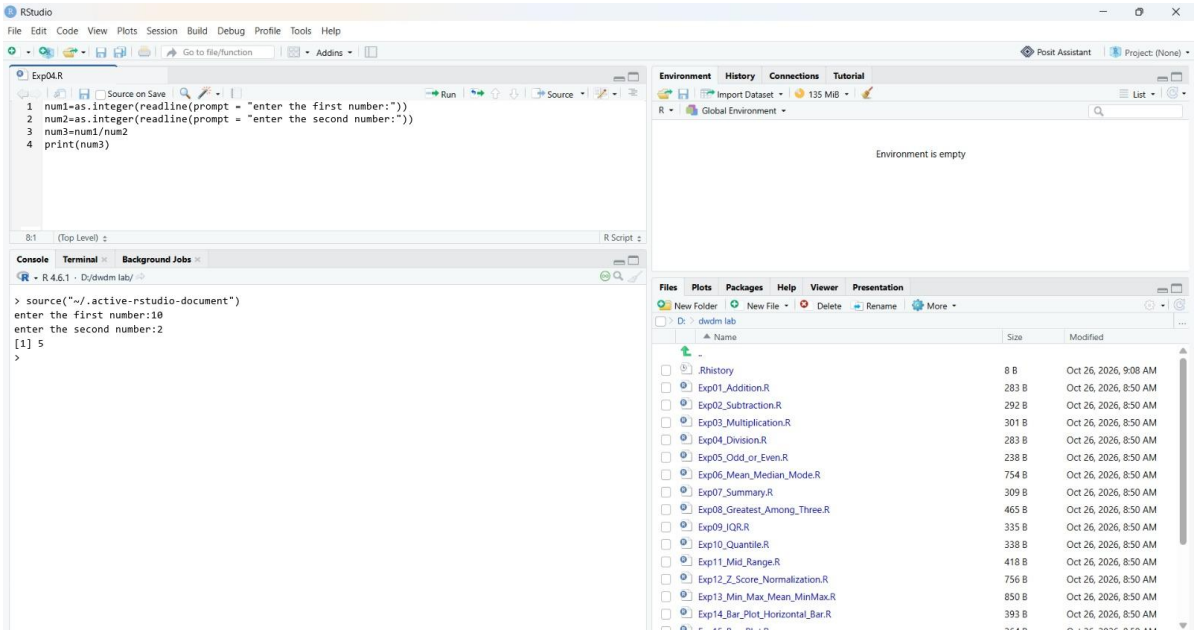
Aim

To prove the program for division using R-tool.

Program

```
num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
num3=num1/num2
print(num3)
```

Output



Result

Thus, the basic program division was executed successfully.

EXPERIMENT 5: ODD OR EVEN

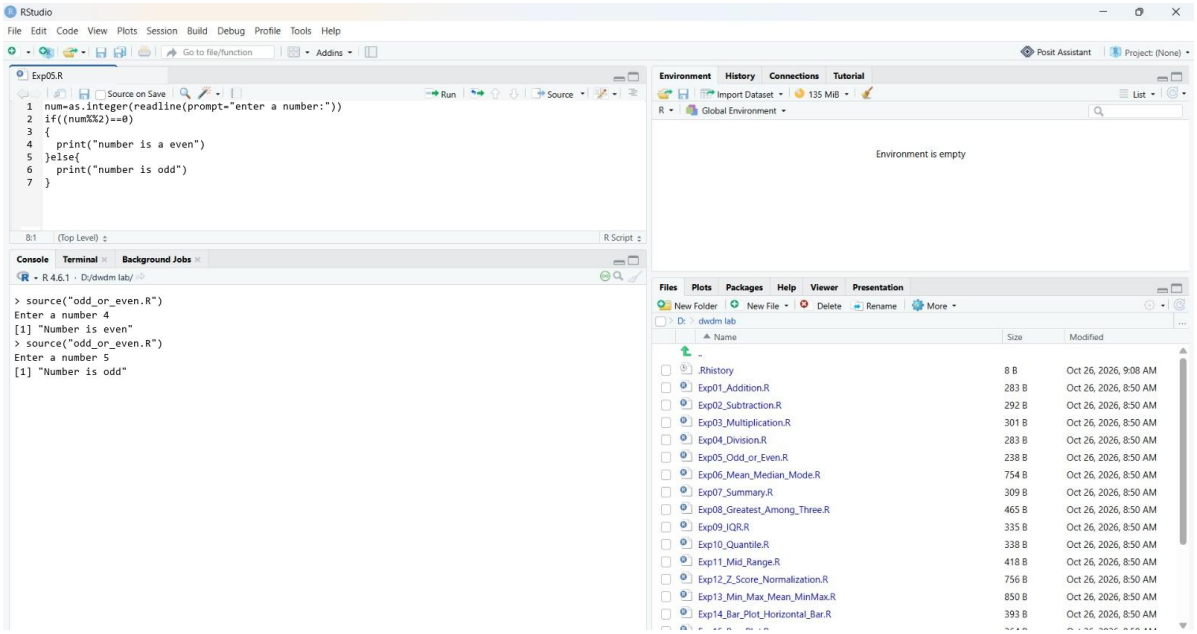
Aim

To write the program for odd or even using R-tool.

Program

```
num=as.integer(readline(prompt="enter a number:"))
if((num%%2)==0)
{
  print("number is a even")
}else{
  print("number is odd")
}
```

Output



Result

Thus, the basic program odd or even was executed successfully.

EXPERIMENT 6: MEAN,MEDIAN,MODE

Aim

To write the program for mean,median,mode.

Program

MEAN

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
mean(df $age)
write.csv(df,"datafr.csv")
```

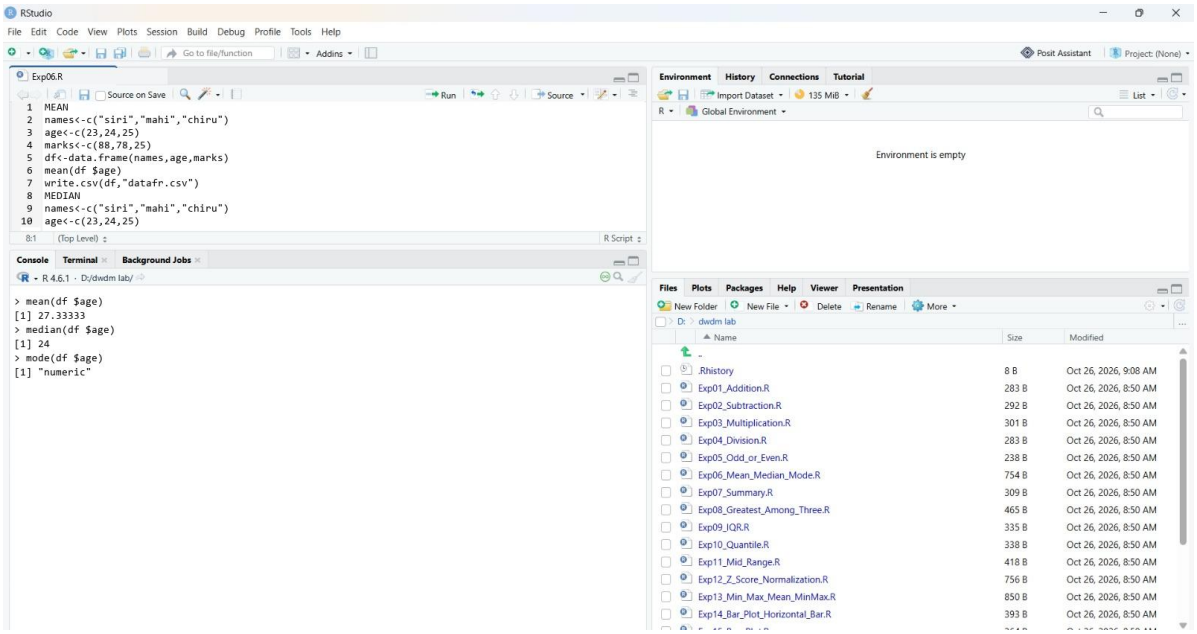
MEDIAN

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
median(df $age)
write.csv(df,"datafr.csv")
```

MODE

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
mode(df $age)
write.csv(df,"datafr.csv")
```

Output



Result

Thus the central tendency and measure of dispersion is executed successfully.

EXPERIMENT 7: SUMMARY

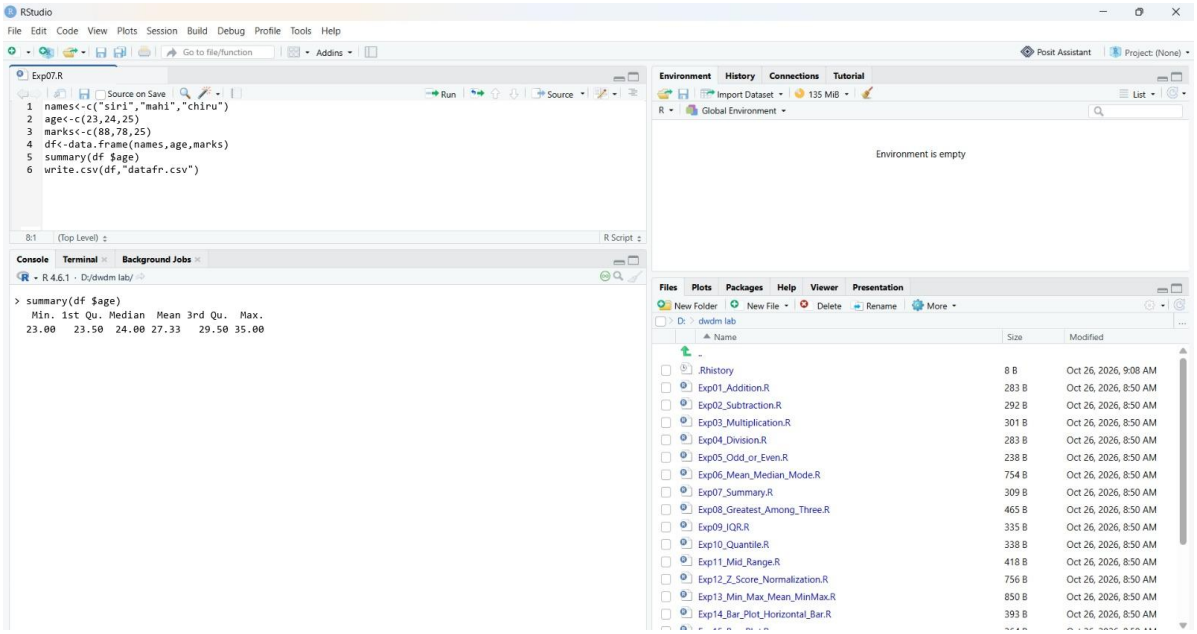
Aim

To write the program for summary using R-tool.

Program

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
summary(df $age)
write.csv(df,"datafr.csv")
```

Output



Result

Thus, the central tendancy and measure of dispersion is executed successfully.

EXPERIMENT 8: GREATER AMONG THREE NUMBERS

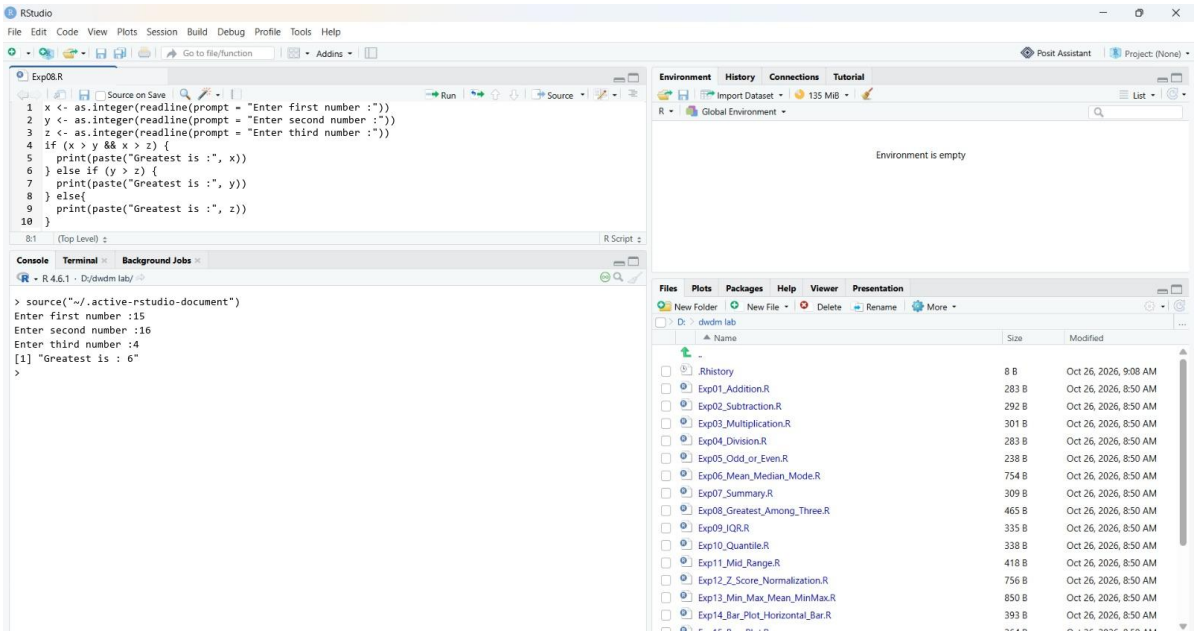
Aim

To write the program for the greatest among three numbers.

Program

```
x <- as.integer(readline(prompt = "Enter first number :"))
y <- as.integer(readline(prompt = "Enter second number :"))
z <- as.integer(readline(prompt = "Enter third number :"))
if (x > y && x > z) {
  print(paste("Greatest is :", x))
} else if (y > z) {
  print(paste("Greatest is :", y))
} else{
  print(paste("Greatest is :", z))
}
```

Output



Result

Thus, the greatest among the three numbers was executed successfully.

EXPERIMENT 9: IQR

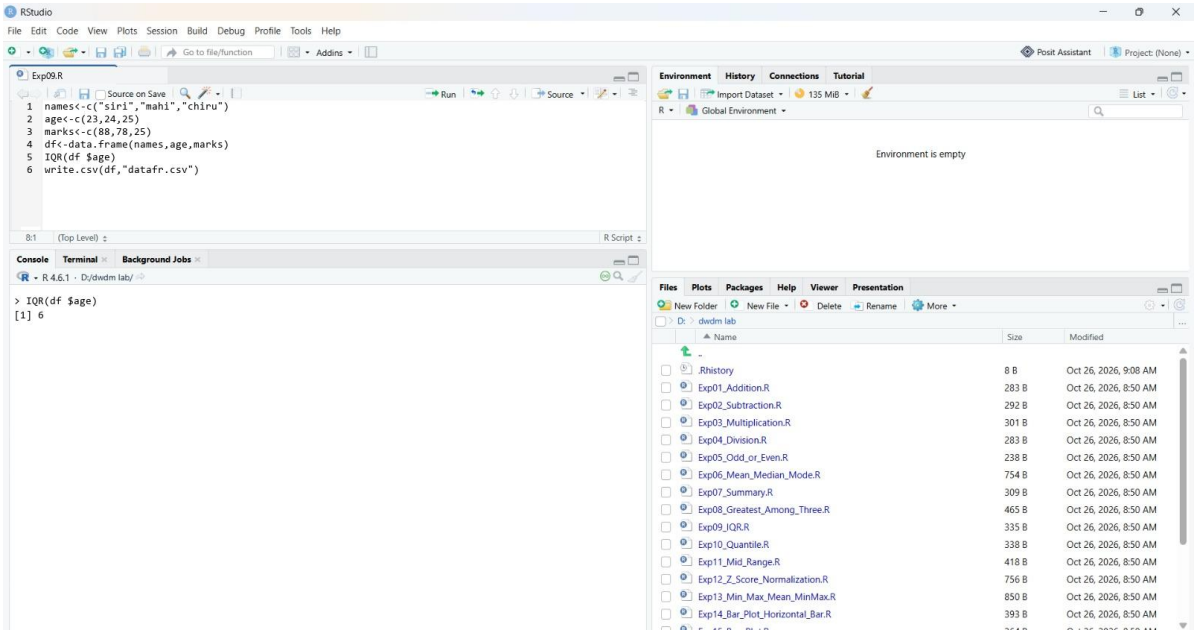
Aim

To write the program for central tendency and data dispersion measures using R tool.

Program

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
IQR(df $age)
write.csv(df,"datafr.csv")
```

Output



Result

Thus, the program for central tendency and data dispersion measures was executed successfully.

EXPERIMENT 10: QUANTILE

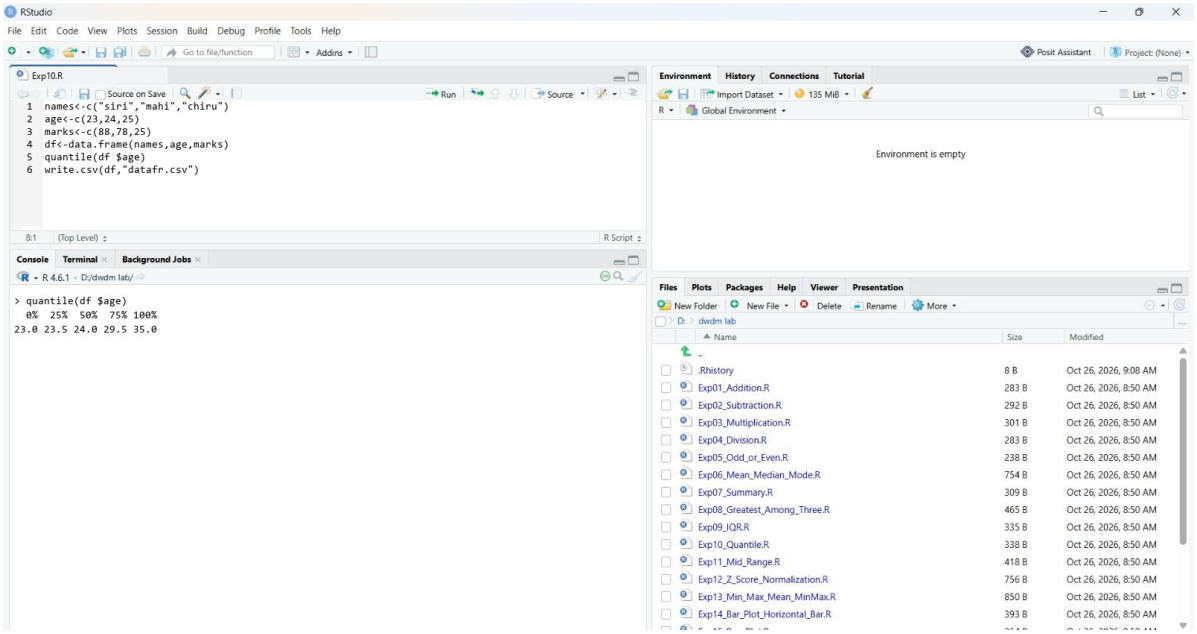
Aim

To write the program for central tendency and data dispersion measures.

Program

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
quantile(df $age)
write.csv(df,"datafr.csv")
```

Output



Result

Thus, the program for central tendency and data dispersion measures was executed successfully

EXPERIMENT 11: MID RANGE

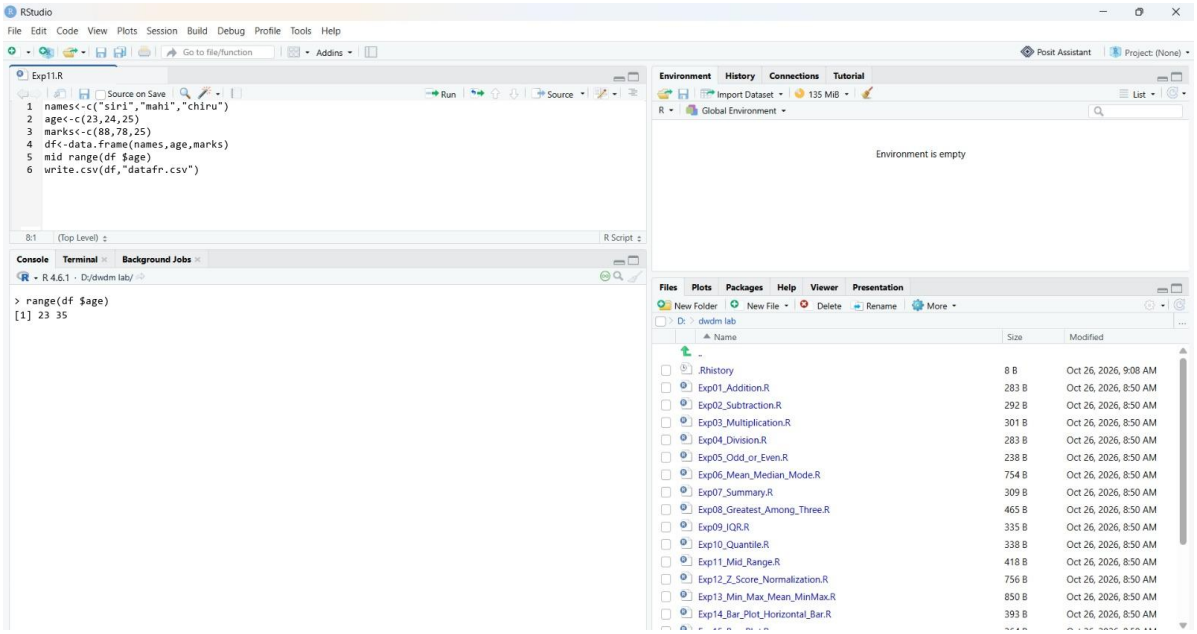
Aim

To write the program for central tendency and data dispersion measures.

Program

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
mid range(df $age)
write.csv(df,"datafr.csv")
```

Output



Result

Thus, the program for central tendency and data dispersion measures was executed successfully

EXPERIMENT 12: Z-SCOORE NORMALIZATION

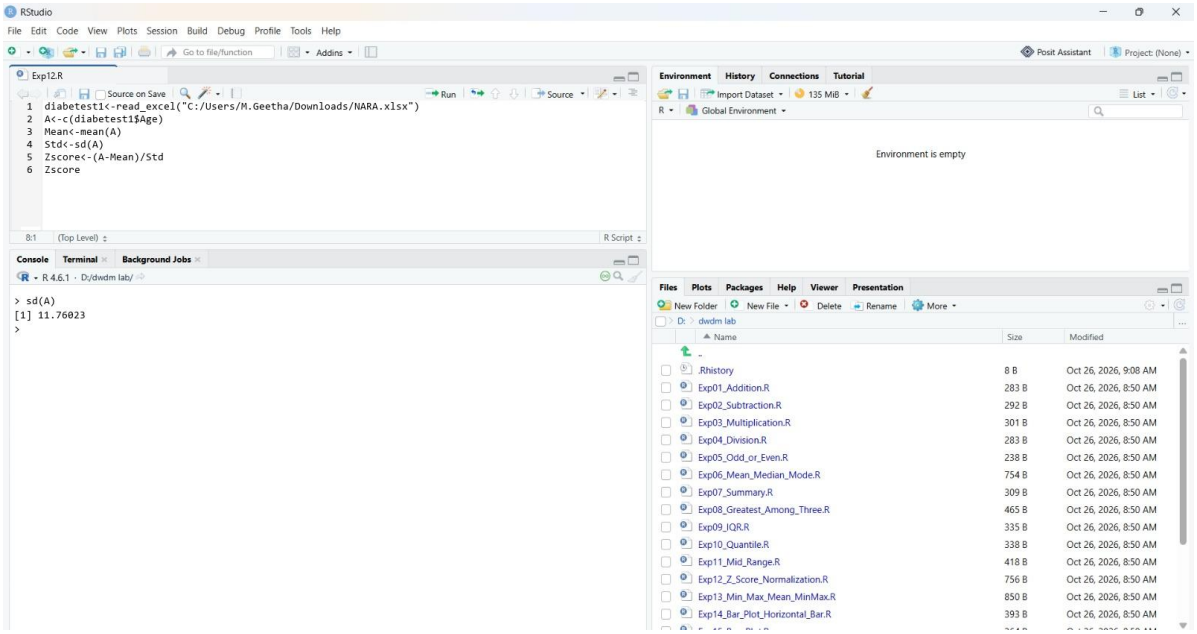
Aim

To write the program for Z-scoore normalization using R-tool.

Program

```
diabetest1<-read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")
A<-c(diabetest1$Age)
Mean<-mean(A)
Std<-sd(A)
Zscore<-(A-Mean)/Std
Zscore
```

Output



Result

Thus, the Z-score normalization using R tool was executed successfully.

EXPERIMENT 13: MIN,MAX,MEAN,MINMAX

Aim

To write the program for the minimum,maximum,mean and minmax using r-TOOL

Program

MEAN

```
diabetest1<-read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")
A<-c(diabetest1$Age)
Mean<-mean(A)
```

MINIMUM

```
diabetest1<-read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")
A<-c(diabetest1$Age)
Minimum<-Min(diabetest1$Age)
```

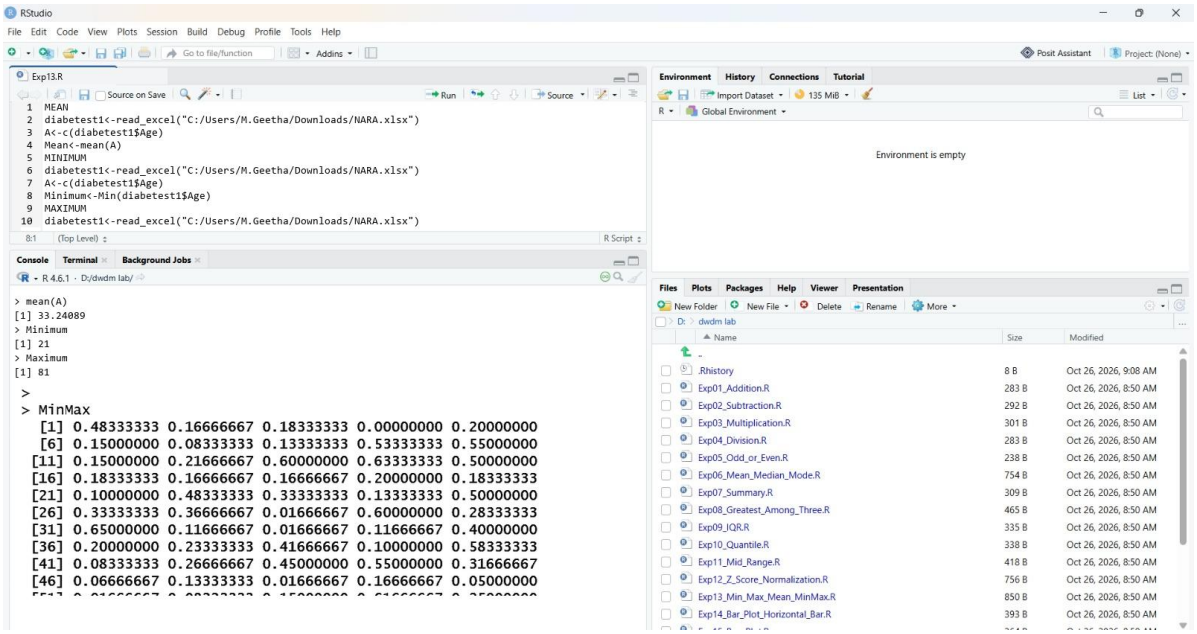
MAXIMUM

```
diabetest1<-read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")
A<-c(diabetest1$Age)
Maximum<-Max(diabetest1$Age)
```

MINMAX

```
diabetest1<-read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")
A<-c(diabetest1$Age)
Maximum<-Max(diabetest1$Age)
Minimum<-Min(diabetest1$Age)
MinMax<-(A-Minimum)/(Maximum-Minimum)
MinMax
```

Output



Result

Thus, the program for min,max,minmax,mean was executed successfully.

EXPERIMENT 14: BAR PLOT AND HORIZONTAL BAR

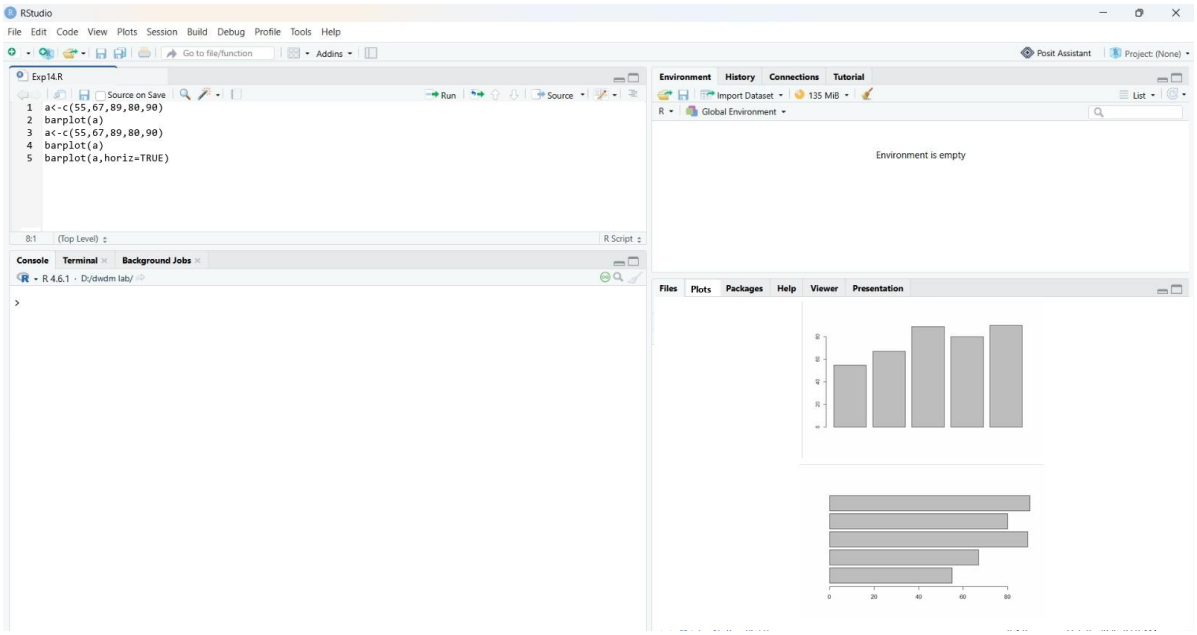
Aim

To draw the bar plot and horizontal bar using R-tool.

Program

```
a<-c(55,67,89,80,90)
barplot(a)
a<-c(55,67,89,80,90)
barplot(a)
barplot(a,horiz=TRUE)
```

Output



Result

Thus, the bar and horizontal bar plot was executed successfully.

EXPERIMENT 15: BOX PLOT

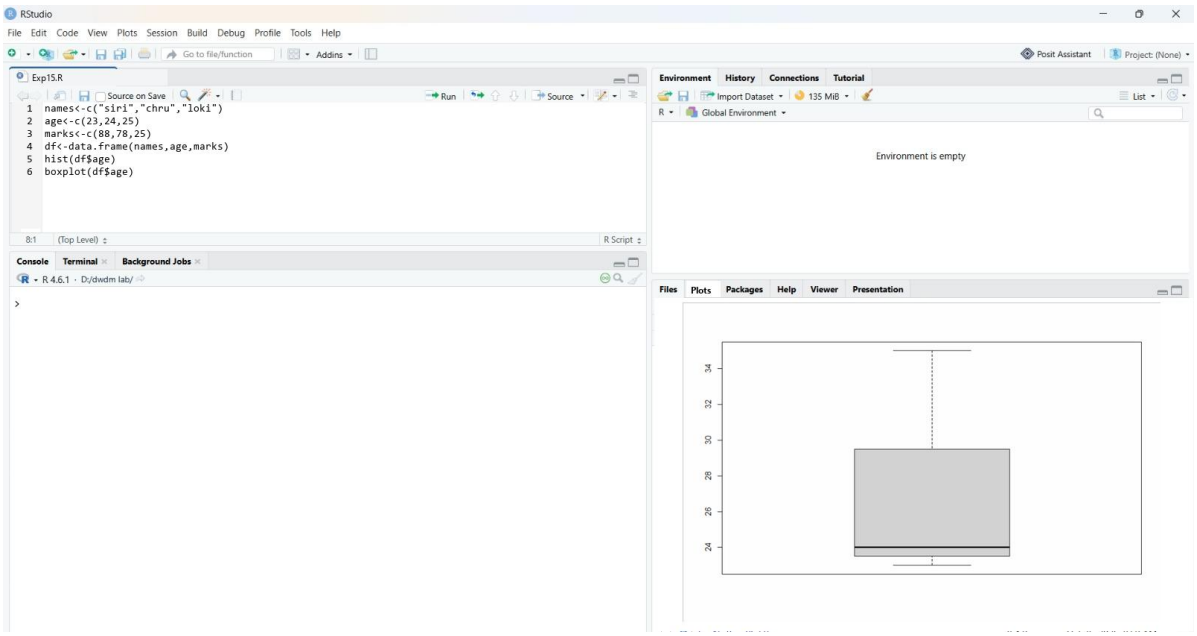
Aim

To draw the box plot using R-tool.

Program

```
names<-c("siri","chru","loki")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
hist(df$age)
boxplot(df$age)
```

Output



Result

Thus, the box plot was executed successfully.

EXPERIMENT 16: HISTOGRAM

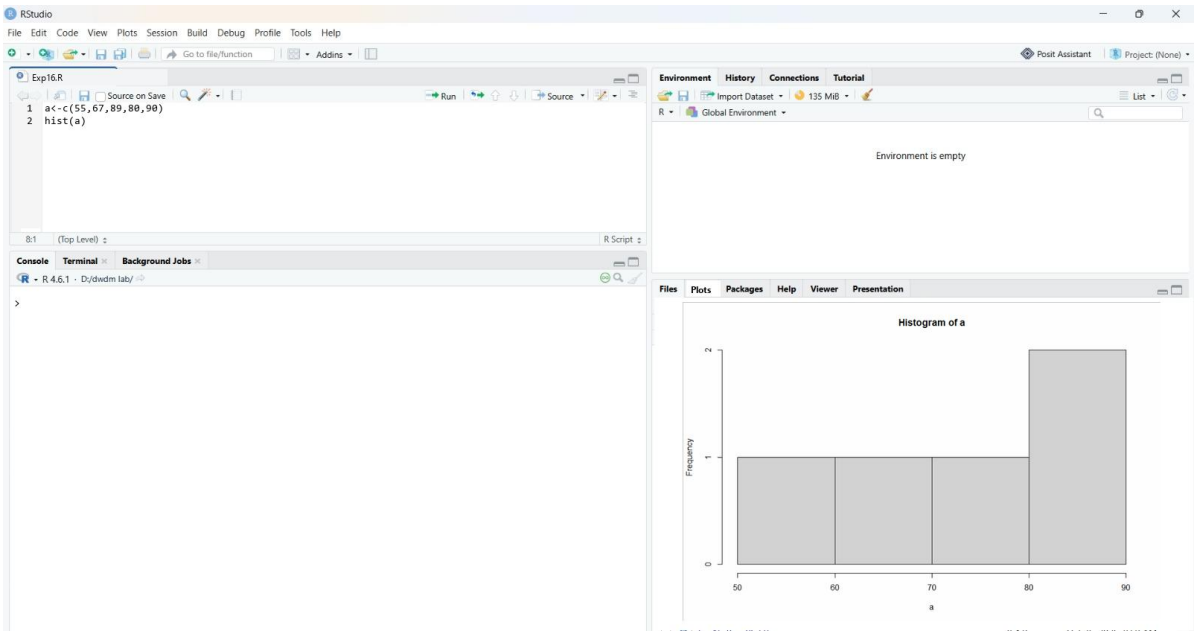
Aim

To draw the histogram plot using R-tooll.

Program

```
a<-c(55,67,89,80,90)
hist(a)
```

Output



Result

Thus, the histogram plot was executed successfully.

EXPERIMENT 17: CORRELATION ANALYSIS

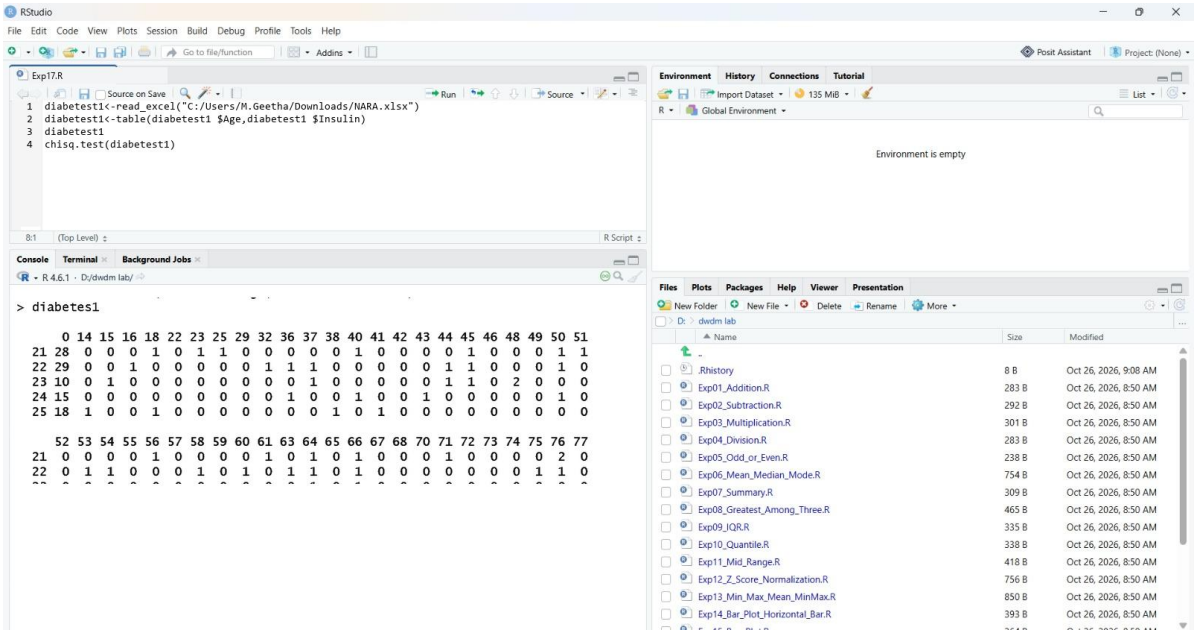
Aim

To write the program for correlation analysis using R-tool.

Program

```
diabetest1<-read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")
diabetest1<-table(diabetest1 $Age,diabetest1 $Insulin)
diabetest1
chisq.test(diabetest1)
```

Output



Result

Thus, the correlation analysis was executed successfully.

EXPERIMENT 18: SCATTER PLOT

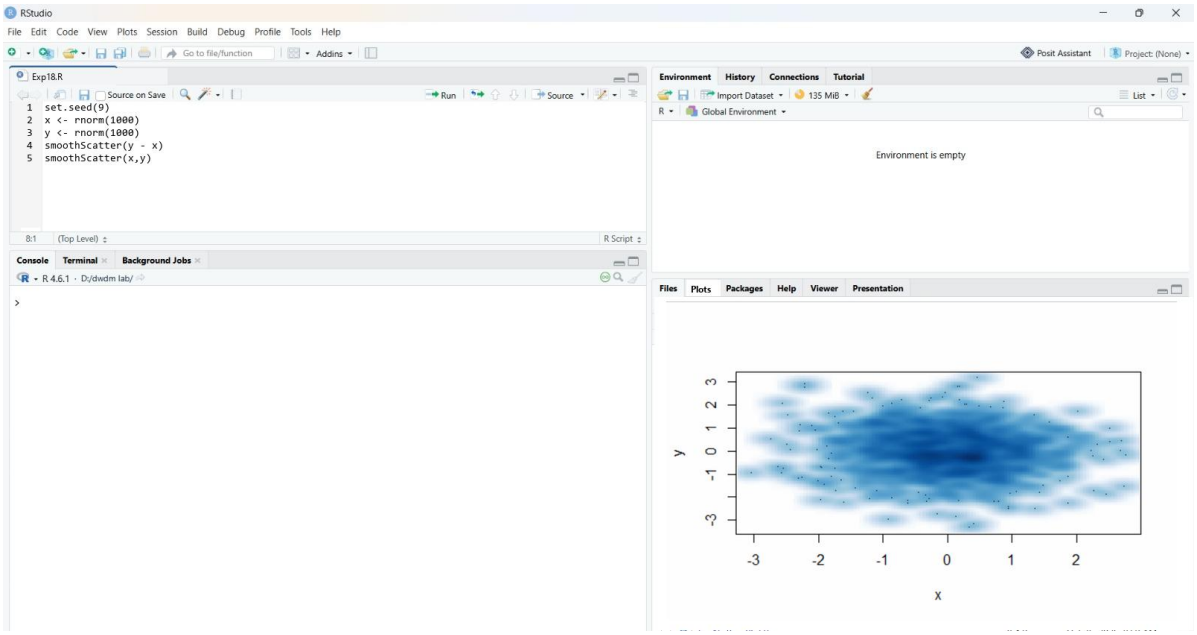
Aim

To draw the scatter plot using R-tool

Program

```
set.seed(9)
x <- rnorm(1000)
y <- rnorm(1000)
smoothScatter(y - x)
smoothScatter(x,y)
```

Output



Result

Thus, the scatter plot was executed successfully.

EXPERIMENT 19: LINEAR REGRESSION

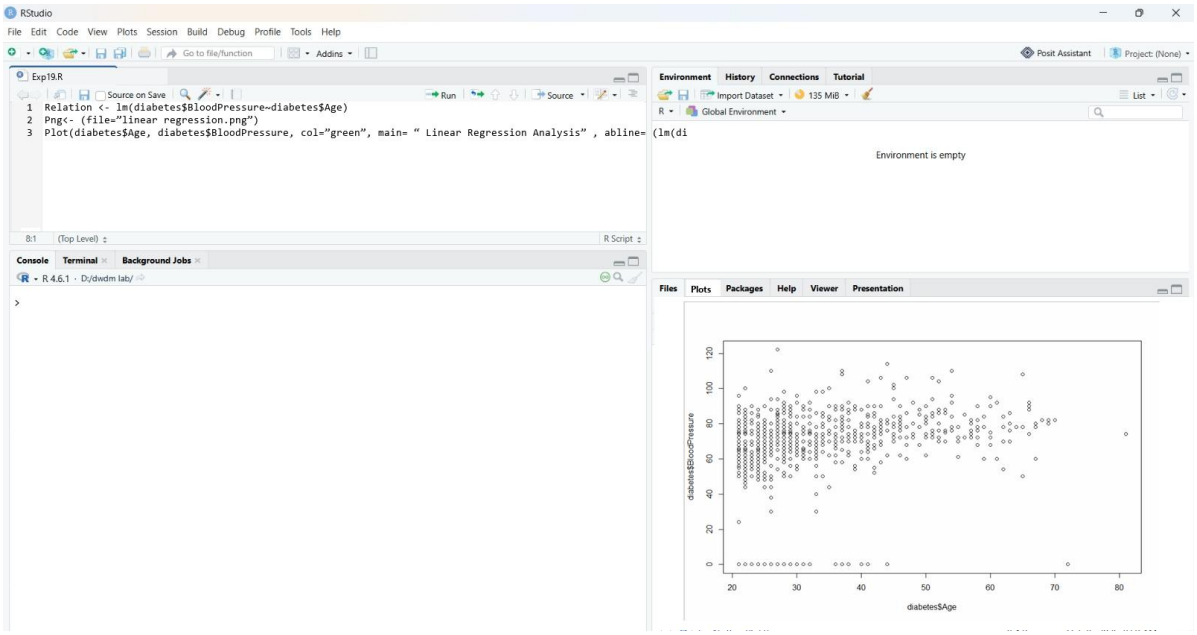
Aim

To write thr program for the linear regression using R-tool.

Program

```
Relation <- lm(diabetes$BloodPressure~diabetes$Age)
Png<- (file=”linear regression.png”)
Plot(diabetes$Age, diabetes$BloodPressure, col=”green”, main= “ Linear
Regression Analysis” , abline= (lm(diabetes$BloodPressure~ diabetes$Age)),
xlab = “BloodPressure”, ylanb= “Age”)
```

Output



Result

Thus, the linear regression program was executed successfully.

EXPERIMENT 20: MULTIPLE REGRESSION

Aim

To write the program for the multiple regression.

Program

```
Input <- diabetes[,c("Age", "BloodPressure", "Glucose")]
Model <- lm(Age~ BloodPressure+Glucose,data=input)
Print(model)
A<- coef(model)[1]
Print(A)
xBloodPressure<- coef(model)[2]
yGlucose<- coef(model)[3]
print(xBloodPressure)
print(yGlucose)
y = A+xBloodPressure + yGlucose
print(y)
```

Output

